PROJECT DESIGN PHASE SPECIFICATION

StaffSync HRMS — Solution Architecture Specification

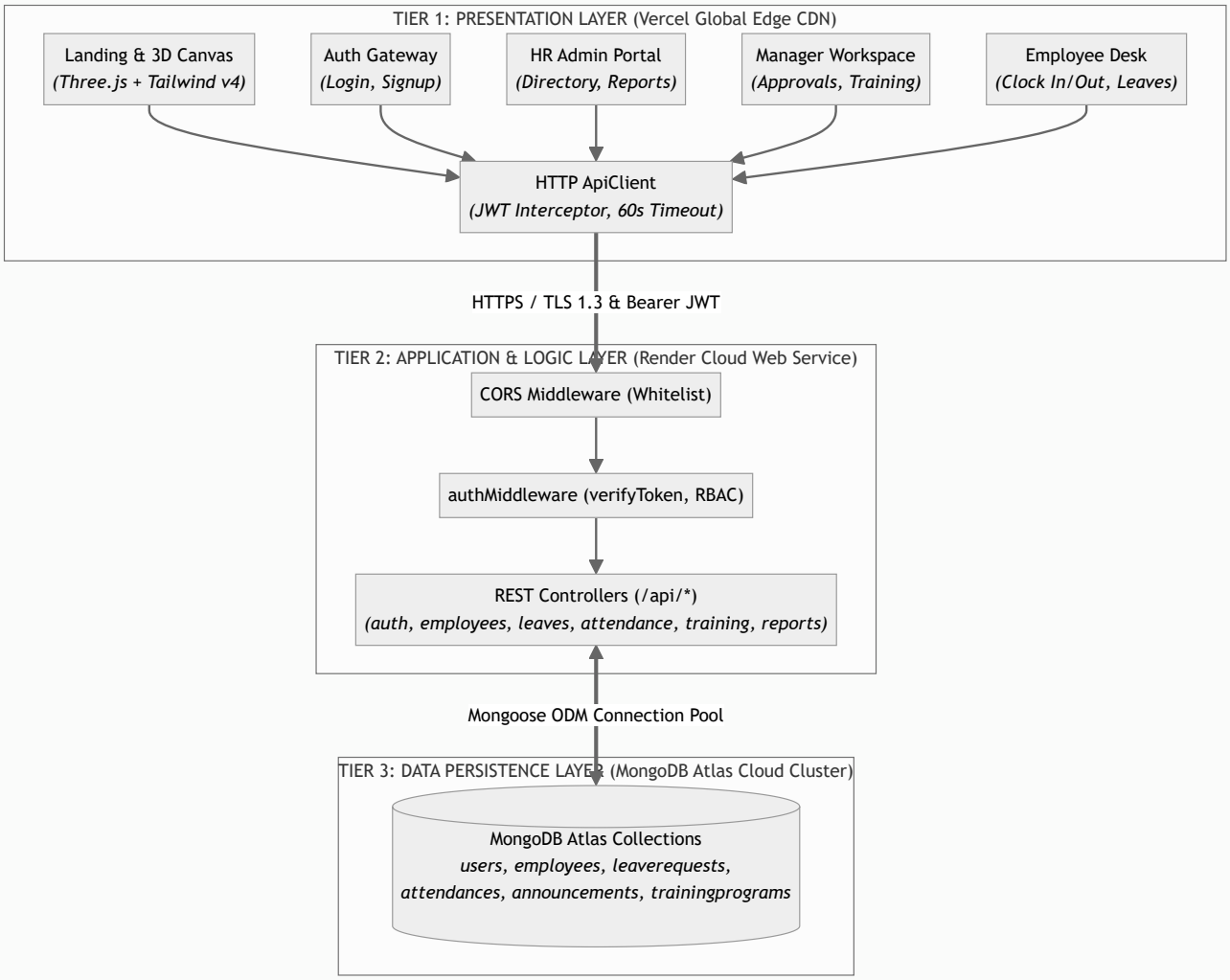
3-tier enterprise cloud systems engineering, security middleware pipelines, and cloud topology.

Project Name	**StaffSync — Enterprise Human Resource Management System**
Architecture Pattern	Decoupled 3-Tier Enterprise Cloud (MVC + SPA)
Version & Lead	v1.0.0 (Production Release) | Parikshit Sharma
Live Endpoints	Client: staffsync-smoky.vercel.app | API: staffsync-czzq.onrender.com

1. Solution Architecture Diagram

The end-to-end cloud topology encompasses presentation edge delivery, containerized API gateways, and distributed cloud document storage:

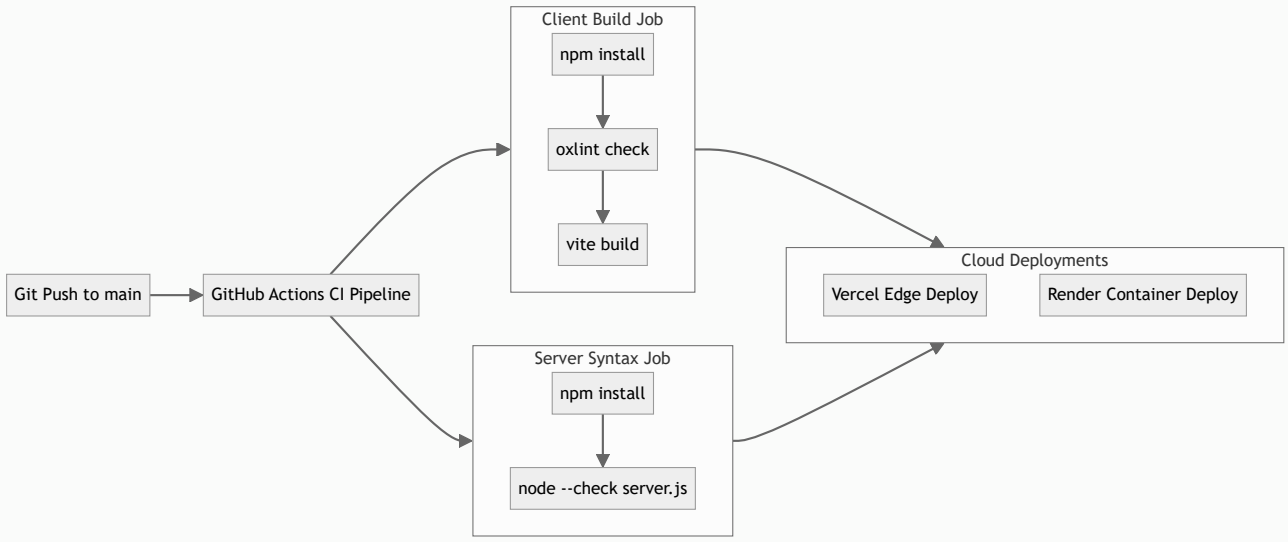
FIGURE 1: HIGH-LEVEL END-TO-END SOLUTION ARCHITECTURE (3-TIER CLOUD)



2. CI/CD Pipeline & Automated Quality Gates

Every commit or pull request to the primary branch initiates automated testing and verification:

FIGURE 2: CONTINUOUS INTEGRATION & DEPLOYMENT PIPELINE FLOW



3. Tier Breakdown & Security Controls

Tier / Domain	Key Technologies	Core Security & Operational Controls
Presentation Tier	React 19, TypeScript, Vite 8, Tailwind v4	Protected client routing, token storage, 60s timeout cold-start resilience.
Application Tier	Node.js 20+, Express 4, Mongoose 8	bcryptjs password hashing (factor 10), stateless JWT verification, CORS whitelist.
Persistence Tier	MongoDB Atlas Cloud Cluster	TLS 1.3 encrypted data in transit, storage encryption at rest, atomic updates.